

Abstract

The emergence of cryptographically relevant quantum computers (CRQC) poses a terminal threat to current public-key infrastructure, specifically RSA and Elliptic Curve Cryptography (ECC). These systems rely on mathematical problems that Shor’s algorithm can solve in polynomial time on a quantum processor — a task computationally infeasible for classical machines — rendering existing digital security foundations obsolete. This threat is an immediate operational crisis: recent research suggests that quantum hardware requiring as few as 10,000 qubits may be sufficient to execute Shor’s algorithm, compressing the projected threat timeline significantly. Furthermore, “Harvest Now, Decrypt Later” (HNDL) attacks — where adversaries capture encrypted data today for future quantum decryption — mean that sensitive assets in government and military infrastructure are already being compromised. In response, the National Institute of Standards and Technology (NIST) finalized Post-Quantum Cryptography (PQC) standards in 2024, centering on ML-KEM (FIPS 203) for key encapsulation and AES-256 for symmetric data protection. However, legacy IoT hardware and resource-constrained devices lack the computational capacity to execute these lattice-based algorithms natively — the increased key sizes and processing cycles exceed the performance limits of standard microcontrollers deployed in critical infrastructure. While software-based edge-server solutions exist, they cannot protect hardware nodes operating in disconnected, remote, or classified environments. This work introduces the PQC Shield: an FPGA-based hardware bridge designed to retrofit legacy devices with quantum-resistant security at the physical layer. Acting as a cryptographic wrapper, the device intercepts communication and applies ML-KEM and AES-256 encryption without requiring infrastructure replacement. This project details the VHDL architecture, implementation, and performance validation of the PQC Shield, presenting a scalable and deployable hardware pathway for securing high-value digital assets in the post-quantum era.